

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #1

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1

Agenda of the Day

- ❑ Welcome!
- ❑ Know your professor
- ❑ Syllabus
- ❑ Introduction to main tools:
 - KUHub Learn, KOLT tutoring

2

Know your professor 😊

- ❑ Name: Ceyda Oğuz
- ❑ Office: ENG 223
- ❑ Phone: X1793
- ❑ E-mail: coguz@ku.edu.tr
- ❑ Background:
 - Highschool – Ankara Atatürk Anadolu Lisesi –
 - Graduated as top scoring student

3

B.S. in IE Middle East Technical University



4

M.S. and Ph.D. in IE Bilkent University



Brought new insights, new opportunities 😊

5

Post-doc – Business School Hong Kong Polytechnic University



Herkes gider Mersin'e ben giderim tersine 😊

6

11 years in Hong Kong (academic life)



7

Since 2004: Professor at Koç University



8

Research Interests

- ▣ Sustainability: Logistics and scheduling



- ▣ Mathematical programming, Metaheuristic algorithms

9

Personal Interests

- ▣ Painting



- ▣ Swimming

10

Syllabus

- ▣ **Lecture hours:** Tu,Th 10:00AM - 11:10AM
- ▣ There will be a Teaching Assistant, a student Teaching Assistant and a KOLT tutor.
- ▣ **Textbook:** Introduction to Industrial and Systems Engineering, *Author:* Turner, W.C., Mize J.H, Case K.E, Nazemetz J.W., *Publisher:* Prentice Hall, 3rd ed. (recommended, optional)

11

Syllabus

- ▣ **Recommended Readings**
 - Yeralan, S., *An Introduction to Industrial Engineering through Computation*, 2016.
 - McNickle, D., Daellenbach, H., *Management Science: Decision-Making Through Systems Thinking*, Palgrave Macmillan, 2005.
 - Fraser, J.M., *Introduction to Industrial Engineering*, online at www.introtoie.com

12

Course Description

- ❑ Introduction to industrial engineering concepts.
- ❑ Fundamentals of systems analysis and modeling.
- ❑ Basics of production and service systems.

- ❑ Several industrial engineering topics with their applications in real life systems, modeling of real-life problems and the solution techniques of these models will be introduced.
- ❑ The course aims to fulfill this task by introducing topics at the introductory level.

13

Main Topics

- ❑ Introduction to Industrial Engineering
- ❑ Production and Service Systems
 - Process planning and management
- ❑ Modeling Concepts and Introduction to Linear Programming
- ❑ Logistics and Supply Chain Management
- ❑ Network Models and Transportation Problems
- ❑ Probability

14

Assessment

- ❑ Midterm exam: 40%
- ❑ Final exam: 40%
- ❑ Homeworks: 15%

- ❑ Exact times and dates of these exams will be announced by the registration office

- ❑ Online modules: 5%

15

Online Modules

- ❑ There are three online modules that you need to complete within this course. After you watch the video for each online module, you will take the online exam. You will get either a “pass” or “fail” from these exams. These modules will contribute 5% of your final grade only if you will pass all online modules. The videos will be uploaded to the KUHub Learn.

16

Online Modules

- ❑ Module 1: Engineering Ethics:
 - Module 1a: Introduction to Ethics consists of two video recordings, which include an introduction to the subject of ethics and a comparison of the concepts of ethics and morality, and a test exam.
 - Module 1b: Ethics from the perspective of engineering consists of a video recording containing the code of ethics on a departmental basis and a test exam.

17

Online Modules

- ❑ Module 2: Engineering Standards includes a video recording of introductory lectures on engineering standards and a test exam.

18

Basic Exam Rules

- ❑ The examinations will be closed-book and closed-notes.
- ❑ Any kind of unauthorized help is strictly forbidden.
- ❑ Violations will be penalized based on University rules (please read the syllabus for details).
- ❑ There will be NO REMEDIAL EXAM. Please see the University guidelines regarding this issue.

19

Course Policies

- ❑ **Homeworks:** There will two homework assignments to be graded.
- ❑ Also study questions will be given as we progress through the topics. These will not be graded.
- ❑ Students can get help from TA for these questions however each student should solve the questions before seeking any help.

20

Course Policies (contd.)

- ❑ **Late policy:** Late homework submission will have the following penalty:
If a homework assignment is submitted within k days later than the due date, the assignment will be evaluated out of $(100-10k)\%$, where k can be at most 3.

21

Course Policies (contd.)

- ❑ **Makeup policy:** Makeup will be given only for the exams and students should have a valid health excuse. The time of makeup exam for midterm will be within one week of the announced exam date. The makeup exam date for final exam will be announced by the University.
- ❑ Students should obey the *Academic Dishonesty Policy of Koç University*.

22

Course Policies (contd.)

□ Code of conduct:

- **Attendance:** It is not compulsory and **will not** affect your grade. However, it has been proven in the past that attending lectures face to face increases the success rate in the course.

24

Course Policies (contd.)

□ Code of conduct:

- **Office Hours:** All students should make use of these times for their questions. If there is a need, TAs and the instructor can be contacted via e-mail outside these office hours and students should expect a reply within 24 hours.

25

Course Policies (contd.)

▣ Code of conduct:

– E-mail communication:

- Subject line must have the format «INDR 100: *main issue*». Writing must be courteous.
- You should expect a reply within 24 hours. Please do not flood us with e-mails before 24 hours of your first e-mail hoping for a quick reply.
- You have to manage your time appropriately.

26

Course Policies (contd.)

▣ Cheating policy

- students caught cheating get **no credit for that assignment**; will be **reported to assoc. dean**
- you may discuss the assignment with others, but **must write up your own solutions**
- **Using AI based sources without clear acknowledgment is also a form of cheating**

▣ Feedback, comments, suggestions, questions are always most welcome.

- Come see me at the office or by e-mail
- Office hours: by appointment
- Office hours of TA on course-website

27

Academic Dishonesty

- ❑ Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct (<http://vpaa.ku.edu.tr/academic/student-code-of-conduct>).
- ❑ Actions considered as academic dishonesty at Koç University include but are not limited to **cheating**, **plagiarism** and **collusion**.
 - Copying from others or providing answers or information, written or oral, to others is cheating.
 - Copying from another student's paper or from another text without written acknowledgment is plagiarism.
 - Unauthorized help from another person or having someone else write one's paper or assignment is collusion.
- ❑ Cheating, plagiarism and collusion are serious offenses resulting in an F grade and disciplinary action.

28

On-line Tools

- ❑ KUHub Learn will be used as the official site of the course. All information about the course will be available there.
- ❑ Students should check there for announcements, assignments, lecture notes, important dates, etc. regularly.
- ❑ **NO** recordings of the lectures will be available for this course.

29

On-line Resources

- ❑ <http://www.iienet.org> (IIE) Institute of Industrial Engineers
- ❑ <http://www.informs.org> (INFORMS) Institute for Operations Research and Management Sciences
- ❑ <http://www.euro-online.org> (EURO) Association of European Operational Research Societies
- ❑ <https://estiem.org/> (ESTIEM) European Students of Industrial Engineering and Management
- ❑ <http://www.ifors.org> (IFORS) International Federation of Operational Research Societies
- ❑ <http://www.yad.org.tr> (YAD) Yöneylem Araştırması Derneği